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Determination of sub-diffraction-limited organelles and biomolecules using microsphere lenses in combination with fluorescence microscopy

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ABSTRACT

We propose the use of a high-refractive index glass microsphere in combination with a conventional fluorescence microscope for imaging sub-cellular organelles and biomolecules. The microsphere is placed on a sample that is immersed in water, collects the near-field nano-features of the sample and generates a magnified virtual image in the far-field, which is recorded through a conventional water immersion objective. We first investigate the imaging capability of the microspheres on size-calibrated fluorescent micro-/nano-particles. The experimental results obtained from a microsphere with 60 μm in diameter demonstrate imaging capability of features of ~λ/7-size (λ is the wavelength) with a magnification factor of 5.4. The position of the virtual image, the field-of-view (FOV) of the microsphere and the magnification factor are studied by using microspheres with different sizes. Finite Element Method (FEM) simulations are performed, providing key insight into the imaging effect of the microspheres with super-resolution capability. Moreover, the distribution and complex shape of different sub-cellular organelles, like centrioles, mitochondria and chromosomes in the AML12 cell line, are imaged with help of the glass microspheres. Thereafter, the subcellular location of mitochondrial encoded proteins could be studied.

Keywords: Fluorescence microscopy, microsphere, near-field, virtual image

1. INTRODUCTION

A conventional optical microscope has a strict limitation in spatial resolution: diffraction makes that the minimum size that can be resolved in a sample is around one half of the wavelength of illumination, which is a few hundred nanometers in the visible (infrared) wavelength region [1]. It is well known that microspheres that are significantly larger than the illumination wavelength can be treated as focusing lenses [2]. When the optically transparent spheres are several to tens of micrometers in size, they can generate a 'photonic nanojet', which exits, upon illumination, from such microspheres with a waist smaller than the diffraction limit [3, 4]. This is at the basis of their superior resolution capability and is key for their imaging potential beyond the classical diffraction limit [5, 6]. The capability of microspheres to image sub-diffraction-limited objects depends on parameters, including the refractive index ratio of the microsphere and its surrounding medium, the illumination wavelength, and the size of the microsphere [7, 8]. By carefully tuning these parameters, the microspheres show the capability of imaging nano-objects with super-resolution. Commercially available microspheres have been used in a number of optical microdevices [8-10]. Especially, microspheres formed from melts yield exquisitely smooth surfaces that provide superior optical characteristics. Moreover, such microspheres are inexpensive and are available in a range of sizes and materials [11]. Here, we present the use of high-refractive index (n_p = 1.92) barium titanate glass microspheres in combination with a conventional fluorescence microscope for imaging of organelles and biomolecules in cells. In the demonstrated technique, the microspheres are simply put on a sample that is immersed in water (refractive index n_m = 1.33), and projects the sample's near-field nano-features into the far-field, generating a magnified virtual image that is recorded through a conventional 40× water immersion objective with numerical aperture (NA) of 0.8, as conceptually illustrated in Figure 1.

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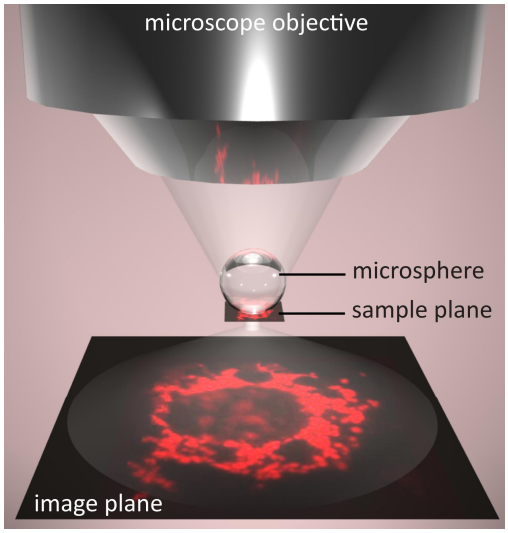

Figure 1. Imaging of organelles and biomolecules by using a microsphere in combination with a conventional fluorescence microscope. A transparent glass microsphere is put on top of cells, which are immersed in water-based medium. A magnified virtual image is generated in the far-field and recorded by a water immersion objective. Sub-diffraction features are resolved thanks to a photonic nanojet that emerges from the high-refractive index microsphere.

2. MICROSPHERE ON SIZE-CALIBRATED FLUORESCENT PARTICLES

To test the imaging effect of the microspheres with conventional fluorescence microscopy, micron-size calibrated polystyrene micro-/nano-particles are firstly used as samples. In the experiments, the microspheres with diameter of 60 µm, as well as the sample are immersed in de-ionized water, and a water immersion objective with NA of 0.8 is used. Figure 2 shows the images obtained from the fluorescent particles with emission wavelength λ_p = 680 nm. Comparing Figure 2(a) and 2(b), a 5.4× magnification factor is demonstrated when using a microsphere with diameter of 60 µm for imaging a 1 µm-diameter particle. Subsequently, 100 nm-diameter particles are dispersed on a glass substrate, which cannot be resolved through conventional fluorescent microscopy (see Figure 2(c)). However, these particles are clearly imaged through the microsphere in Figure 2(d). Therefore, the fluorescence experiments demonstrate an imaging capability of an object of ~ λ/7-size by using the 60 µm microsphere with the water immersion objective.

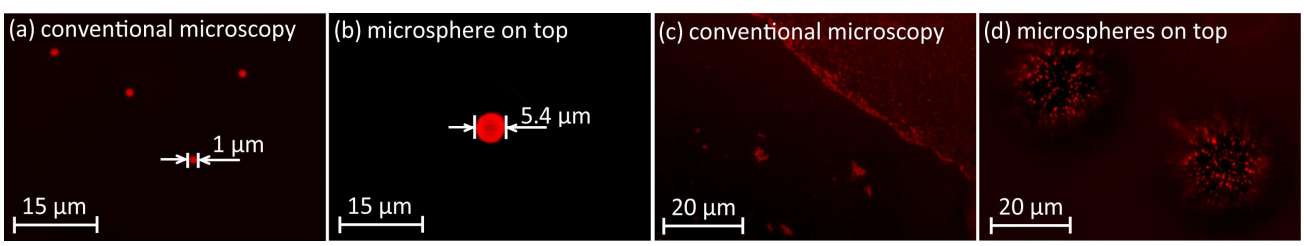

Figure 2. Comparison of conventional microscopy and use of a 60 µm microsphere for imaging fluorescent micro-/nano-particles: (a,c) conventional microscopy on 1 µm and 100 nm particles and (b,d) using a microsphere.

Afterwards, the position of the virtual image and the FOV of the microsphere are investigated. Here, we define the image plane on which the virtual image shows the strongest fluorescent signal as the best image plane, and the distortion-free area in the center of the virtual image as the FOV of the microsphere. We use microspheres with diameter from 40 µm to 100 µm to image 200 nm fluorescent particles. From the experimental results, the distance between the best image plane and the rear surface of the microsphere increases with the size of the microspheres (see Figure 3(a)). Moreover, a linear dependence of the FOV on the size of the microspheres is also observed, as shown in Figure 3(b).

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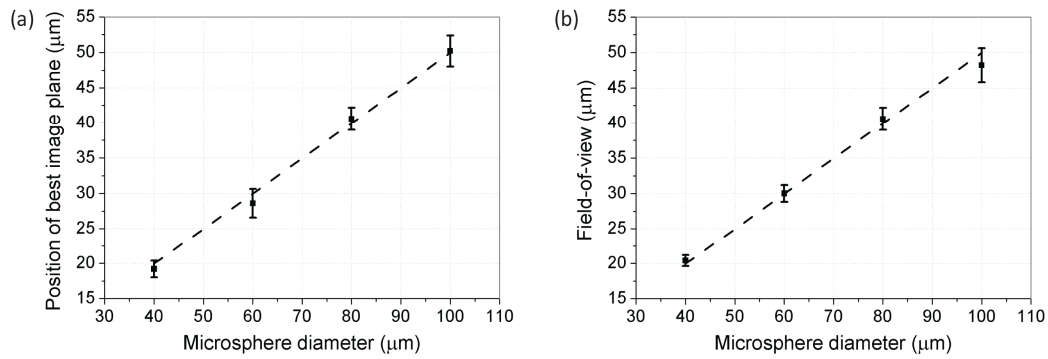

Figure 3. Measurement of microspheres with different sizes on 200 nm fluorescent particles. The experiment shows that (a) the position of the best image plane and (b) the FOV of the microsphere linearly increase with the size of the microspheres. The data are obtained from 50 measurements, points represent the average and error bars the variance, while dashed lines are guides to the eye.

Moreover, the magnification factor of the virtual image is studied by using microspheres with diameter between 2 μm and 100 μm to image 200 nm fluorescent particles, as shown in Figure 4. When the microsphere is smaller than 6 μm, the magnification factor increases with the microsphere diameter. The magnification factor reaches the maximal value ~ 5.8 when the size of the sphere is between 6 μm and 8 μm. Then the magnification factor decreases with the sphere size. In order to explain the experimental results, we perform a FEM study of the electromagnetic wave propagation through the microsphere (n_p = 1.92) and its surrounding water medium (n_m = 1.33). The numerical study is carried out using a scalar equation to study the electromagnetic field distribution in the vicinity of glass microspheres with different sizes in COMSOL Multiphysics software. Figure 5 shows simulation results of the optical intensity distribution obtained from microspheres with diameter of 2 μm, 5 μm, 10 μm and 20 μm, respectively, verifying that the microspheres can focus the illumination into a small region with highly enhanced optical field, producing a 'photonic nanojet'. The photonic nanojet is situated on the rear surface of the 5 μm sphere (see e.g. Figure 5(b)) where the sample is located. Therefore, the best imaging quality, corresponding to the biggest magnification factor, should be obtained from the 5 μm sphere. This conclusion basically matches the experimental results. When the sphere is 2 μm in size, the photonic nanojet merges into the microsphere, therefore, the imaging capability of the microsphere is highly hampered. If the microsphere is bigger than 5 μm, the photonic nanojet starts moving away from the sphere's rear surface. Therefore, the magnification decreases along with the increased sphere size. By comparing the experimental results with the analysis described above, the origin of the super-resolution is clearly pointed out: (i) the development of the photonic nanojet by the microsphere, and (ii) the microsphere also acting as a solid immersion lens and providing local enhancement of the refractive index and a reduction of the effective wavelength. On a fundamental level, the strongly reduced transverse dimension of the photonic nanojet directly indicates the super-resolution capability of the microsphere.

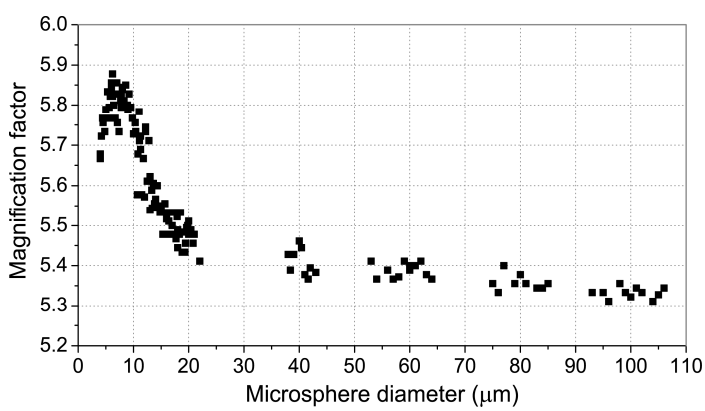

Figure 4. Magnification factor of images that are obtained from microspheres with different sizes. Dots are results obtained from experiments.

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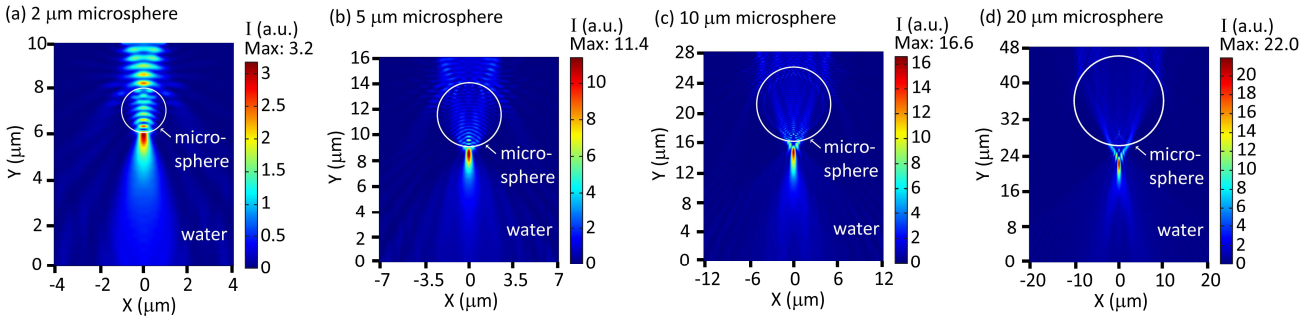

Figure 5. FEM simulation of light propagating through a microsphere which is immersed in water. The light intensity distribution is studied for microspheres with diameter of (a) 2 μm, (b) 5 μm, (c) 10 μm and (d) 20 μm, respectively. The result clearly indicates a photonic nanojet generated by the microsphere which is illuminated by a plane wave.

3. MICROSPHERE ON ORGANELLES AND BIOMOLECULES

Since the microsphere shows strong capability to resolve much smaller objects than the diffraction limit in water with a classical fluorescence microscope setup, it makes this technique suitable for imaging in life sciences research. To investigate the performance of the microsphere on biological samples, we perform the experiments on several different organelles from cells. In particular, the mouse hepatocyte cell line AML12 (alpha mouse liver 12) is used. The centrioles in the cells are stained by rabbit polyclonal Ab to γ-tubulin (1st Ab) and DyLight® 649-conjugated goat polyclonal Ab to rabbit IgG (2nd Ab). The mitochondria and chromosomes in the cells are stained by MitoTracker® Deep Red FM probe and 4', 6-diamidin-2-phenylindol (DAPI), respectively. The cells are fixed and preserved in phosphate buffered saline (PBS) solution and imaged through a 60 μm microsphere with the water immersion objective. We first image centrioles, which form the core of the centrosome and are primarily composed of γ-tubulin proteins. In eukaryotic cells, centriole cylinder-like structures are built of nine triplet microtubules and have approximate diameters of 150 – 200 nm and lengths of 400 – 450 nm [12, 13]. Each centrosome comprises two centrioles, of which, the older one, frequently referred to as the mother, is perpendicular to the other [14]. Conventional optical microscopy can only reveal the presence of centrioles but not their sub-structure, as shown in Figure 6(a) and 6(c). When a microsphere nanoscope is placed on top of the centriole(s), the ring-like cross-section of the centriole is clearly revealed (see Figure 6(b)). Moreover, the link between the mother and the daughter centriole is imaged (see Figure 6(d)). Thereafter, stained mitochondria and chromosomes are imaged using the water immersion objective without and with the microsphere (see Figure 7), the latter can better resolve their complex shapes. Actually, the 40× objective used in the experiments can provide a large FOV, many microspheres can be observed simultaneously. Therefore, a population of nano-objects can be imaged at the same time. This holds the capability to cover a large sample area, providing representative images with statistical relevance, which is important for any biological study.

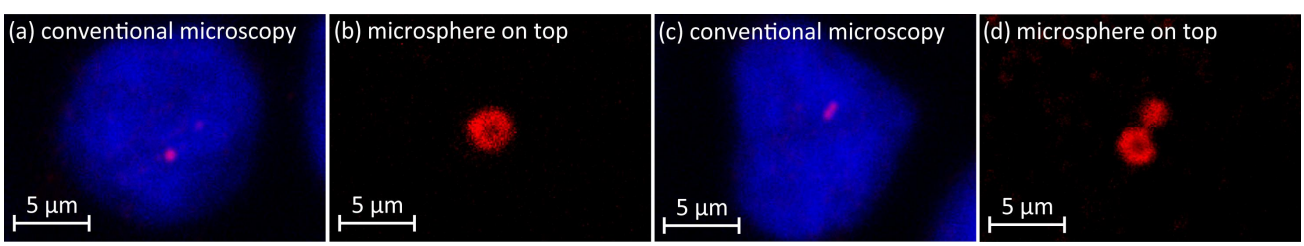

Figure 6. Comparison of conventional fluorescent microscopy and the use of a microsphere on centrioles. (a) γ-tubulin protein on a single centriole (in red) is detected as a faint spot close to nucleus (in blue), but in (b) the ring structure is clearly resolved. (c,d) γ-tubulin detected on a mother and daughter centriole, the link between the two centrioles being clearly observed through the microsphere.

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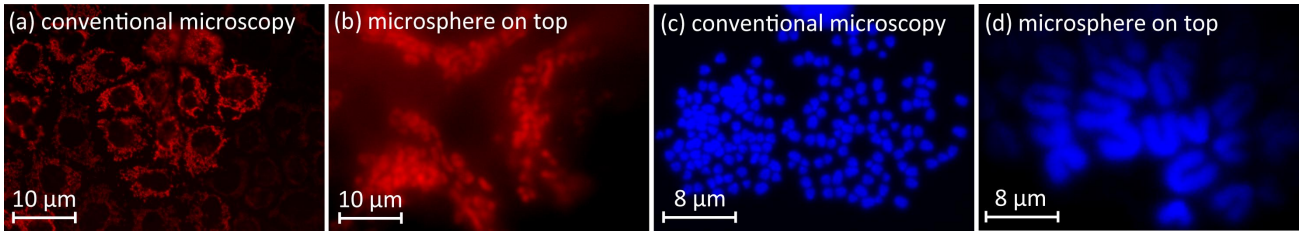

Figure 7. Fluorescent microscopy of (a,b) mitochondria and (c,d) chromosomes. Comparison of (a,c) conventional fluorescent microscopy and (b,d) the use of a microsphere nanoscope.

Moreover, we study the imaging with the microsphere of mitochondrial encoded protein MTCO1 in AML12 cells. The MTCO1 protein is part of respiratory complex IV, the subcellular location of MTCO1 is closely related to its function and constitutes an essential aspect for understanding the complex machinery of living cells. The AML12 cells are treated with doxycycline or with dimethyl sulfoxide (DMSO) as control. Figure 8 shows that the level of MTCO1 decreases robustly upon the doxycycline treatment. The conventional optical microscope cannot provide details on the localization of the MTCO1 neither can identify differences in the distribution of MTCO1 upon doxycycline treatment due to the diffraction limit. However, when a 60 µm glass microsphere is placed on top of the sample, the state of localization of MTCO1 protein is enabled.

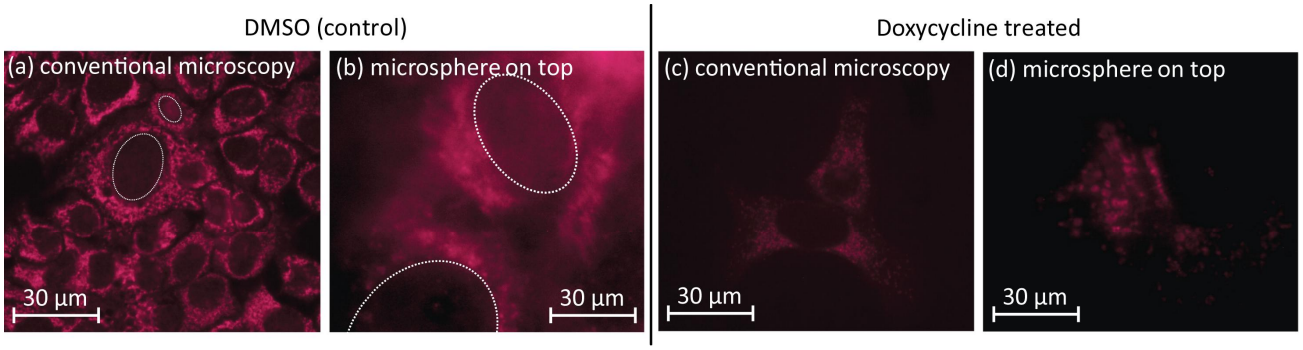

Figure 8. Fluorescent images showing the distribution of MTCO1 proteins using (a,c) a conventional microscopy and (b, d) with additional use of a microsphere. Compared to the control experiments with DMSO shown in (a, b), doxycycline treatment diminishes MTCO1 expression, as shown in (c, d). The black ellipsoidal areas in (a, b) are nuclei (some have been indicated by dotted lines), in which there is no MTCO1 expression; the latter only exists in the cytoplasm.

4. CONCLUSIONS

In conclusion, we achieve super-resolution imaging of nanostructures by using a high refractive index glass microsphere in combination with a conventional water immersion objective. The fluorescence experiments performed with a 60 µm sphere show that an object with a size as small as ~ $\lambda/7$, i.e. about one fifth of the resolution of the microscope objective, can be quantitatively determined, while a 5.4× magnification of the image is obtained. The distribution of the fluorescent particles can be imaged through the 60 µm sphere in a distortion-free area of several 10 square microns. The distance from the virtual image to the microsphere surface and the FOV show a linear dependence on the size of the microspheres. The experimental results also show that the magnification of the image varies with the size of the microspheres and reaches a maximal value of 5.8 when a 6 microns microsphere is used. The variation on the magnification is mainly due to the different position of the photonic nanojet emerged from microspheres with different sizes. This conclusion is also verified by the FEM simulations. The photonic nanojet with waist smaller than the diffraction limit is key for the microsphere-enabled high resolution imaging that goes beyond the Rayleigh criterion of the microscope objective. Moreover, different sub-cellular organelles, i.e. centrioles, mitochondria and chromosomes in the AML12 cell line, are imaged through the glass microsphere. Not only the sub-structure of the centrioles can be resolved, which is beyond the limit of imaging with a normal microscope objective, but also the distribution and complex shape of mitochondria and chromosomes can be imaged through the 60 µm sphere in combination with a 40× water

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immersion objective. The imaging method is also used to identify the expression of the specific mitochondrial membrane protein MTCO1. We think that in future, due to the straightforwardness of this approach, this in turn provides new pathways for super-resolution optical imaging in real time, which is useful for life science applications and optical nano-sensing.

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